

Frontend

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Frontend Development

Vite

Vite is a modern front-end build tool created by Evan You, the original author of Vue.js. It is designed to provide a faster and more efficient development experience for web applications, particularly those built with modern front-end frameworks such as Vue, React, and Preact.

Vite's key feature is its use of native ES modules, which allows it to serve code directly to the browser without bundling or transpiling. This results in faster development server start-up times and instant hot module replacement (HMR) during development.

To learn more about Vite, [check this out](#)

Quick Start

Let us use Vite to create a simple React Project.

- Ensure that you are on Node.js v18+ or 20+

- 1. Create the Project using the following command:
` ` `
`pnpm create vite`
` ` `

- 2. Follow the prompts
` ` `
- ✓ Project name: ... csc01-project
- ✓ Select a framework: > React
- ✓ Select a variant: > TypeScript + SWC
- ` ` `

- 3. Open the Project
` ` `
- `cd csc01-project`
 ` ` `

- 4. Setup and Run the Project
` ` `
- `pnpm install`
 `pnpm dev`
 ` ` `

To learn more about Vite Templates, [check this out](#)

Frontend Development

Web Frameworks

As web applications become increasingly complex, the need for efficient and effective development tools has never been greater. JavaScript frameworks provide developers with the necessary resources to create dynamic and responsive user interfaces. With so many options available, it can be overwhelming to choose the proper framework. Let's look closely at some of the leading JavaScript frameworks and what they bring to the table.

React

Developed by Facebook, React is a library for building user interfaces, particularly single-page applications. Its component-based architecture allows developers to create reusable UI components, making code more manageable and scalable. React's virtual DOM enhances performance by minimizing direct interactions with the actual DOM, leading to faster rendering.

1. **Component-Based Architecture:** React's modular approach allows developers to create reusable components that manage their state and logic, leading to cleaner and more maintainable code.
2. **Virtual DOM:** React uses a Virtual DOM to optimize performance. Changes are first made to the Virtual DOM, which is then efficiently reconciled with the real DOM, minimizing direct manipulation and enhancing user experience.
3. **Declarative Syntax:** Declarative syntax allows developers to describe the UI in a given state, making the code more intuitive and easier to understand.
4. **JSX:** JSX is a syntax extension that allows developers to write HTML-like code within JavaScript. It improves readability and makes it easier to visualize the UI structure.
5. **State Management:** React provides built-in state management, allowing components to manage their state. For more complex applications, libraries like Redux and Context API can be integrated for global state management.
6. **Rich Ecosystem and Community:** React has a vast ecosystem of libraries and tools and strong community support, making it easy for developers to find resources and solutions.

To learn how to use React, [check this out](#).

Frontend Development

Next.js

Next.js has rapidly gained popularity among developers as a robust framework for building React applications. Created by Vercel, Next.js simplifies the development process while offering many features that enhance performance, scalability, and user experience.

1. **Server-Side Rendering (SSR):** Next.js supports server-side rendering, allowing pages to be rendered on the server for improved performance and SEO. This results in faster initial load times and better crawlability by search engines.
2. **Static Site Generation (SSG):** With SSG, Next.js can generate static pages at build time and serve them directly from a CDN. This is ideal for content-heavy sites, ensuring quick load times while allowing dynamic content through APIs.
3. **API Routes:** Next.js enables developers to create API endpoints within the same codebase, simplifying backend development and integrating databases or third-party services.
4. **Automatic Code Splitting:** The framework automatically splits code into smaller bundles, loading only the necessary code for the current page, which enhances performance and reduces initial load times.
5. **Image Optimization:** Next.js optimizes images for different devices and screen sizes, ensuring efficient loading without sacrificing quality.
6. **File-Based Routing:** With Next.js, routing is intuitive. Each file in the `pages/app` directory corresponds to a route, making it easy to manage the application structure.

To learn how to use Next.js, [check this out](#).

Frontend Development

Remix

Remix is an innovative framework built on React that aims to enhance the developer experience while providing powerful features for building web applications. With its focus on performance, user experience, and simplicity, Remix is quickly gaining traction among developers.

1. **Nested Routing:** Remix allows for nested routes, enabling developers to create complex layouts and UI structures easily. This feature promotes better organization of components and enhances code maintainability.
2. **Data Loading:** Remix simplifies data fetching by allowing developers to load data at the route level. This allows each route to specify its data requirements, leading to more efficient data management and improved performance.
3. **Optimized for Performance:** Remix is designed with performance in mind. It leverages server-side rendering (SSR) and static site generation (SSG) to ensure fast load times and a smooth user experience.
4. **Form Handling:** Remix provides a robust form-handling system that simplifies managing form submissions and validations. This makes it easier to create interactive and user-friendly forms.
5. **Error Boundaries:** The framework includes built-in error boundaries, which allow developers to handle errors at different application levels gracefully. This feature improves user experience by preventing entire applications from crashing due to unhandled errors.

To learn about Remix, [check this out](#).

Frontend Development

Preact

Preact is a fast, lightweight alternative to React. It offers a similar API but is significantly smaller, making it an excellent choice for developers who want to build high-performance web applications without the overhead of more extensive libraries.

1. **Small Size:** Preact is only about 3KB (gzipped), making it one of the smallest libraries for building user interfaces. This compact size helps improve load times and overall performance, especially on mobile devices.
2. **React Compatibility:** Preact provides a similar API to React, allowing developers to transition between the two easily. Most React components can be used in Preact with little modification, making it a seamless alternative for existing React projects.
3. **Fast Rendering:** Preact is designed for speed, with a virtual DOM implementation that optimizes rendering performance. This ensures that updates to the UI are handled efficiently, resulting in a smoother user experience.
4. **No Dependencies:** Preact is built with a focus on minimalism and does not rely on external dependencies. This reduces project complexity and helps maintain a clean codebase.
5. **Server-Side Rendering (SSR):** Preact supports server-side rendering, allowing developers to render components on the server and send fully rendered HTML to the client. This improves SEO and reduces the time to paint, enhancing the overall user experience.

Frontend Development

Angular

Angular, developed and maintained by Google, is a robust front-end web application framework that has gained immense popularity among developers for building dynamic and responsive web applications. With its rich set of features and strong community backing, Angular simplifies the development process and enhances the performance of web applications.

1. **Component-Based Architecture:** Angular uses a component-based architecture, allowing developers to create reusable components that encapsulate their logic, templates, and styles. This modularity enhances code maintainability and reusability.
2. **Two-Way Data Binding:** With two-way data binding, changes in the user interface are automatically reflected in the data model and vice versa. This synchronization simplifies development and reduces the need for additional code.
3. **Dependency Injection:** Angular's built-in dependency injection system promotes a clean separation of concerns, making applications more modular and testable. Developers can easily manage dependencies, leading to better maintainability.
4. **Routing and Navigation:** Angular's powerful routing module enables the creation of single-page applications (SPAs) with smooth navigation between views without reloading the page, enhancing user experience.
5. **Rich Ecosystem and Tooling:** The Angular CLI streamlines development by providing commands for creating, building, and testing applications. Angular also integrates well with various third-party libraries.
6. **Comprehensive Testing Support:** Angular is designed for testability, making writing unit and end-to-end tests easy using tools like Jasmine.

To learn how to use Angular, [check this out](#).

Frontend Development

Vue.js

Vue.js is a popular JavaScript framework created by Evan Yu in 2014. It has gained significant traction among developers for building user interfaces and single-page applications. Its simplicity and flexibility make it an excellent choice for beginners and experienced developers.

1. **Reactive Data Binding:** Vue.js utilizes a reactive data binding system that allows developers to create dynamic interfaces effortlessly—the view updates automatically when the underlying data changes, ensuring a seamless user experience.
2. **Component-Based Architecture:** Vue's component-based architecture encourages a modular approach to development. This allows developers to encapsulate functionality and styles within reusable components, promoting code reusability and maintainability.
3. **Virtual DOM:** Vue.js employs a virtual DOM to optimize rendering performance. Vue can efficiently update only the changed UI parts by keeping a lightweight representation of the actual DOM, resulting in faster rendering and improved performance.
4. **Directives:** Vue provides a set of built-in directives (such as v-if, v-for, and v-bind) that enable developers to manipulate the DOM declaratively. This makes creating dynamic and interactive applications easy without writing extensive JavaScript code.

To learn how to use Vue.js, [check this out](#).

Frontend Development

Nuxt.js

Nuxt.js is a progressive framework built on Vue.js that simplifies the development of server-side rendered applications and static websites. It provides robust features that enhance the development experience and improve application performance.

1. **Server-Side Rendering (SSR):** Nuxt.js enables server-side rendering out of the box, improving SEO and enhancing applications' initial load time. This feature allows pages to be rendered on the server and sent to the client as fully rendered HTML.
2. **Static Site Generation (SSG):** Nuxt.js makes it easy for developers to generate static websites. This feature is handy for content-heavy sites, as it allows for faster loading times and better performance by serving pre-rendered pages.
3. **File-Based Routing:** Nuxt.js simplifies routing by using a file-based system. Developers can create pages by adding Vue files to the pages directory, and Nuxt.js automatically generates the corresponding routes, streamlining the development process.
4. **Modular Architecture:** Nuxt.js supports a modular architecture, allowing developers to extend the framework's functionality with modules. Numerous community modules are available for features like authentication, analytics, and PWA support, making it easy to enhance applications.
5. **Automatic Code Splitting:** Nuxt.js automatically splits the code for each page, which helps reduce the initial load time and improves performance. This feature ensures that only the necessary code is loaded for each page, optimizing the user experience.

To learn more about Nuxt.js, [check this out](#).

Frontend Development

Svelte

Svelte is a revolutionary JavaScript framework created by Rich Harris in 2016 that has gained popularity for its unique approach to building user interfaces. Unlike traditional frameworks that rely on a virtual DOM, Svelte compiles components into highly efficient imperative code at build time. This results in faster performance and a more streamlined development experience.

1. **No Virtual DOM:** Svelte eliminates the need for a virtual DOM by compiling components into optimized JavaScript at build time. The framework directly manipulates the DOM, leading to faster updates and improved performance.
2. **Reactive Programming:** Svelte embraces a reactive programming model, allowing developers to write less boilerplate code. State changes automatically trigger updates in the UI, making it easier to manage the application state.
3. **Built-in State Management:** Svelte provides a simple and effective way to manage a state without needing external libraries. The built-in store system allows easy state sharing across components, streamlining the development process.
4. **Built-in Transitions and Animations:** Svelte has built-in support for transitions and animations, allowing developers to create smooth and engaging user experiences without relying on external libraries.

To learn how to use Svelte, [check this out](#).

Frontend Development

SvelteKit

SvelteKit is a robust framework for building web applications using Svelte, a modern JavaScript framework that compiles components into highly efficient vanilla JavaScript. With its focus on performance and developer experience, SvelteKit is quickly gaining popularity among developers.

1. **File-Based Routing:** SvelteKit simplifies routing by using a file-based system. Developers can create routes by adding files to the `src/routes` directory, making it intuitive and easy to manage.
2. **Server-Side Rendering (SSR):** SvelteKit supports server-side rendering, allowing for faster initial page loads and improved SEO. This feature enables developers to deliver fully rendered HTML to the client, enhancing user experience.
3. **Static Site Generation (SSG):** Besides SSR, SvelteKit can generate static sites, making it an excellent choice for blogs, portfolios, and documentation sites. This feature allows for pre-rendering pages at build time, resulting in faster load times.
4. **API Routes:** SvelteKit allows developers to create API endpoints directly within the application. This streamlines the development process by enabling you to handle server-side logic and data fetching without needing a separate backend.

To learn how to use SvelteKit, [check this out](#).

Frontend Development

Astro

Astro is a powerful static site generator that has gained popularity for its unique approach to building fast, optimized websites. It allows developers to create content-rich sites with minimal effort while leveraging the latest web technologies. Here are six standout features of Astro:

1. **Partial Hydration:** Astro's innovative partial hydration feature allows developers to load only the necessary JavaScript for interactive components, resulting in faster page loads and improved performance. This allows static content to be served quickly while still enabling dynamic functionality where needed.
2. **Framework Agnostic:** Astro supports multiple front-end frameworks, including React, Vue, Svelte, and Preact. This flexibility allows developers to choose their preferred tools and libraries, making integrating existing projects easier or leveraging the strengths of different frameworks.
3. **Markdown and MDX Support:** Astro natively supports Markdown and MDX, enabling content creators to write in a familiar format while seamlessly integrating components. This feature is handy for blogs and documentation sites, where content and interactivity often go hand in hand.
4. **Built-in Image Optimization:** Astro includes built-in image optimization capabilities, automatically resizing and serving images in modern formats like WebP. This ensures images are delivered quickly and efficiently, enhancing the overall user experience.
5. **Static Site Generation (SSG):** Astro excels at static site generation, allowing developers to pre-render pages at build time. This results in faster load times and improved SEO, as search engines can easily crawl and index static content.
6. **Zero JavaScript by Default:** One of Astro's most compelling features is its zero JavaScript by default philosophy. Astro will not include any JavaScript in the final build unless explicitly specified, leading to faster load times and a more efficient site.

To learn how to use Astro, [check this out](#).

Frontend Development

TypeScript

TypeScript is a superset of JavaScript that adds static typing to the language. Developed and maintained by Microsoft, it allows developers to define types for variables, function parameters, and return values. This can help catch errors at compile time rather than at runtime, enhancing code quality and maintainability, especially in large codebases.

TypeScript compiles down to plain JavaScript, which means it can run in any environment that supports JavaScript, such as web browsers and Node.js. It also supports modern JavaScript features and provides additional tools for object-oriented programming, such as interfaces and generics.

Key features of TypeScript include:

1. **Static Typing:** Developers can specify types for variables, function parameters, and return values, which helps catch type-related errors early in the development process.
2. **Type Inference:** TypeScript can automatically infer types based on the assigned values, reducing the need for explicit type annotations in many cases.
3. **Interfaces and Type Aliases:** These allow developers to define custom types and enforce contracts within their code.
4. **Generics:** TypeScript supports generics, enabling developers to create reusable components that work with various data types.
5. **Tooling and IDE Support:** TypeScript offers excellent tooling support, including autocompletion, navigation, and refactoring capabilities in popular IDEs and text editors.

To learn more about TypeScript, [check this out](#)

Prettier

Prettier is a code formatter that supports various programming languages, including JavaScript, TypeScript, HTML, CSS, and more. Its primary goal is to enforce a consistent style across your codebase, making it easier to read and maintain. Prettier eliminates debates over code style by adhering to predefined rules, allowing teams to collaborate more effectively.

To learn more about Prettier, [check this out](#).

Frontend Development

ESLint

ESLint is a popular open-source tool for identifying and fixing problems in JavaScript code. It enforces coding standards and best practices to help developers maintain code quality and consistency. ESLint analyzes code for potential errors, stylistic issues, and other problems, providing warnings or errors based on predefined rules.

Key features of ESLint include:

1. **Customizable Rules:** Developers can configure ESLint to enforce specific coding styles and practices by enabling or disabling rules according to their project's needs.
2. **Plugins:** ESLint supports plugins that add additional rules or functionality, allowing for more specialized linting based on frameworks or libraries (e.g., React, Vue).
3. **Integration:** ESLint can be integrated into various development environments, including text editors and build systems, making it easy to run linting checks as part of the development workflow.
4. **Automatic Fixing:** ESLint can automatically fix specific issues in the code, saving developers time and effort.

Useful Plugins

- [eslint-config-prettier](#): Turns off all unnecessary rules that might conflict with Prettier.
- [eslint-plugin-jsx-a11y](#): Static AST checker for accessibility rules on JSX elements using [AxeCore](#)
- [typescript-eslint](#): Enables ESLint to run on TypeScript code
- [eslint-plugin-compact](#): Lint the browser compatibility of your code
- [eslint-plugin-jsdoc](#): JSDoc specific linting rules for ESLint.

Frameworks

- [angular-eslint](#): Enables ESLint to run on Angular projects
- [eslint-plugin-astro](#): ESLint plugin for Astro component
- [eslint-plugin-react](#): Enables ESLint to run on React projects
- [Eslint-plugin-react-hooks](#): ESLint plugin enforces the [Rules of Hooks](#) from React.
- [eslint-plugin-svelte](#): Enables ESLint to run on Svelte projects
- [eslint-plugin-vue](#): Enables ESLint to run on Vue.js projects

Frontend Development

(Optional) Browserlist

[Browserslist](#) is a tool that allows developers to specify which browsers and versions their web applications should support. Developers can define supported browsers using a simple configuration format, which helps optimize performance and maintain a consistent user experience across different browsers. Configuration can be done in `package.json`, a `.browserslistrc` file, or build tool settings.

(Optional) UI Frameworks

A UI framework is a collection of pre-written code, libraries, and tools that help developers build user interfaces more efficiently. These frameworks often come with a set of components, such as buttons, forms, and navigation bars, that can be easily customized and integrated into web applications. By using a UI framework, developers can save time, ensure consistency, and improve the overall user experience.

To learn more about this, [check this out](#)